On this issue, the jury is still out. Einstein, Podolsky, and Rosen reasoned that the only sensible explanation for how measurements could reveal that widely separated particles had identical properties was that the particles possessed those definite properties all along (and, by virtue of their common past, their properties were correlated). Decades later, Bell's analysis and Aspect's data proved that this intuitively pleasing suggestion, based on the premise that particles always have definite properties, fails as an explanation of the experimentally observed nonlocal correlations. But the failure to explain away the mysteries of nonlocality does not mean that the notion of particles always possessing definite properties is itself ruled out. The data rule out a local universe, but they don't rule out particles having such hidden properties.

In fact, in the 1950s Bohm constructed his own version of quantum mechanics that incorporates both nonlocality *and* hidden variables. Particles, in this approach, always have both a definite position and a definite velocity, even though we can never measure both simultaneously. Bohm's approach made predictions that agreed fully with those of conventional quantum mechanics, but his formulation introduced an even more brazen element of nonlocality in which the *forces* acting on a particle at one location depend instantaneously on conditions at distant locations. In a sense, then, Bohm's version suggested how one might go partway toward Einstein's goal of restoring some of the intuitively sensible features of classical physics—particles having definite properties—that had been abandoned by the quantum revolution, but it also showed that doing so came at the price of accepting yet more blatant nonlocality. With this hefty cost, Einstein would have found little solace in this approach.

The need to abandon locality is the most astonishing lesson arising from the work of Einstein, Podolsky, Rosen, Bohm, Bell, and Aspect, as well as the many others who played important parts in this line of research. By virtue of their past, objects that at present are in vastly different regions of the universe can be part of a quantum mechanically entangled whole. Even though widely separated, such objects are committed to behaving in a random but coordinated manner.

We used to think that a basic property of space is that it separates and distinguishes one object from another. But we now see that quantum